GD&T Symbols and Guidelines Cheat Sheet

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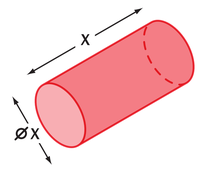
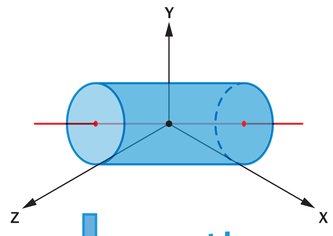
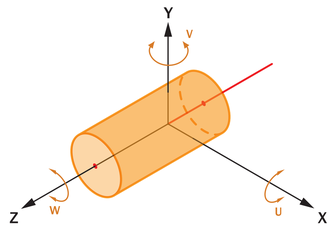
The 4 Fundamental Elements of GD&T:



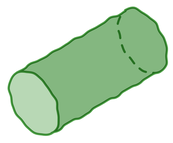
Size



Location



Orientation



Form

All drawings made in first angle projection

	SYMBOL	NAME	ON DRAWING	TOLERANCE ZONE	GAUGING	SURFACES	FEATURES OF SIZE
FORM	—	**Straightness** (Surface)	⌖ 0.03	Two parallel lines 0.03 apart		F	F (Derived Median Line and Plane)
	— Ⓜ	**Straightness** (Derived Median Line w/ Ⓜ)	Ø 10±0.050 ⌖ Ø 0.03 Ⓜ	Cylindrical Tolerance Zone 0.03 when Part Ø = 10.05 0.13 when Part Ø = 9.95	Cylinder Gauge ID = Ⓜ + ⌖ Gauge ID = 10.08		
	▱	**Flatness** (Surface)	⌖ 0.03	Two parallel planes 0.03 apart			
	▱ Ⓜ	**Flatness** (Derived Median Plane w/ Ⓜ)	10±0.05 ⌖ 0.03 Ⓜ	Planar Tolerance Zone Two parallel planes 0.03 apart	Gauge spacing at Virtual Condition Ⓜ + ⌖ = 10.08		
	○	**Circularity**	○ 0.03	Two concentric circles 0.03 apart			
	⌭	**Cylindricity**	⌭ 0.03	Two concentric cylinders 0.03 apart			
ORIENTATION	//	**Parallelism**	// 0.03 A	Two parallel planes 0.03 apart parallel to datum	Flat datum block (Datum A) Gauge kept parallel to datum	OF	O
	⊥	**Perpendicularity** (Feature)	⊥ 0.03 A	Two parallel planes 0.03 apart	Flat datum block (Datum A) Gauge kept perpendicular to datum		
	⊥ Ⓜ	**Perpendicularity** (Feature of Size w/ Ⓜ)	Ø 10±0.05 ⊥ Ø 0.03 Ⓜ A	Axis Tolerance Zone at Ⓜ	Gauge pin inserted perpendicular to datum. Pin Gauge OD = Ⓜ − ⊥ Pin Gauge Ø = 9.92 (9.95 - 0.03) *Multiple datums may be used to specify Perpendicularity*		
	∠	**Angularity**	60° ∠ 0.03 A	Two parallel planes 0.03 apart	Gauge Block		
PROFILE	⌒	**Profile of a line**	⌒ 0.03 A B	Uniform Linear Boundary About True Profile 0.03 apart	Gauge must follow true profile. Profile is usually measured with a CMM.	SLOF	
	⌓	**Profile of a surface**	⌓ 0.03 A B	Uniform Surface Boundary About True Profile 0.03 apart	Gauge must follow true profile. Profile is usually measured with a CMM.		
LOCATION	⌖	**Position**	Ø 10±0.05 ⌖ Ø 0.03 A B C	Part (actual) position; True center; Ø 0.03	Measure X and Y location and compare to the true position. $2X\sqrt{(Actual\ X - True\ X)^2 + (Actual\ Y - True\ Y)^2}$ This formula must be less than the Ø True Position tolerance	None	LO
	⌖ Ⓜ	**Position** (Maximum Material Condition)	Ø 10±0.05 ⌖ Ø 0.03 Ⓜ A B C	Actual Pin Position; True position; Tolerance Zone of Hole Gauge; Gauge Hole Ø = Virtual Condition (Largest Pin Size [MMC] + Position Tol.)	FUNCTIONAL GAUGING: For an External Feature / For an Internal Feature. Tolerance Zone of Pin Gauge; True position; Actual Hole Position; Gauge Pin Ø = Virtual Condition (Smallest Hole Size [MMC] - Position Tol.)		
	◎	**Concentricity**	◎ Ø 0.03 A	Measured axis; Ø 0.03; Datum axis; Datum A; Tolerance Zone	The following is usually done with a CMM: 1. Determine Datum axis 2. Measure referenced surface 3. Determine if central axis falls in TZ	Distribution/ Evenness of Form Only	LOF (Derived Median Points)
	=	**Symmetry**	⌯ 0.03 A	Two parallel planes 0.03 apart	Datum A Plane. The following is usually done with a CMM: 1. Determine Datum plane 2. Measure both surfaces of features 3. Determine if midpoints fall in TZ		
RUNOUT	↗	**Runout**	↗ 0.03 A	Datum A Fixed axially + Rotated	Datum A Fixed + Rotated. *Can also be used on flat surfaces perpendicular to the datum axis*	LOF	
	⌰	**Total Runout**	⌰ 0.03 A	Datum A Fixed axially + Rotated	Datum A Fixed + Rotated. *Can also be used on flat surfaces perpendicular to the datum axis*		

Basic dimensions (not shown) are required to define the True Profile when datums are used.

Diameter Symbol is required

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